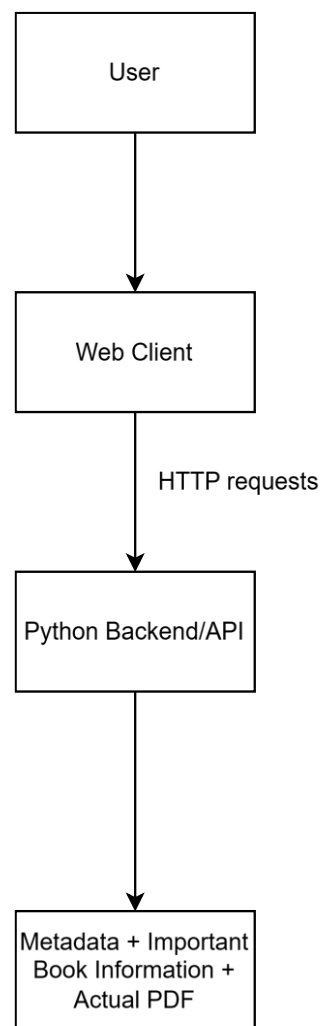
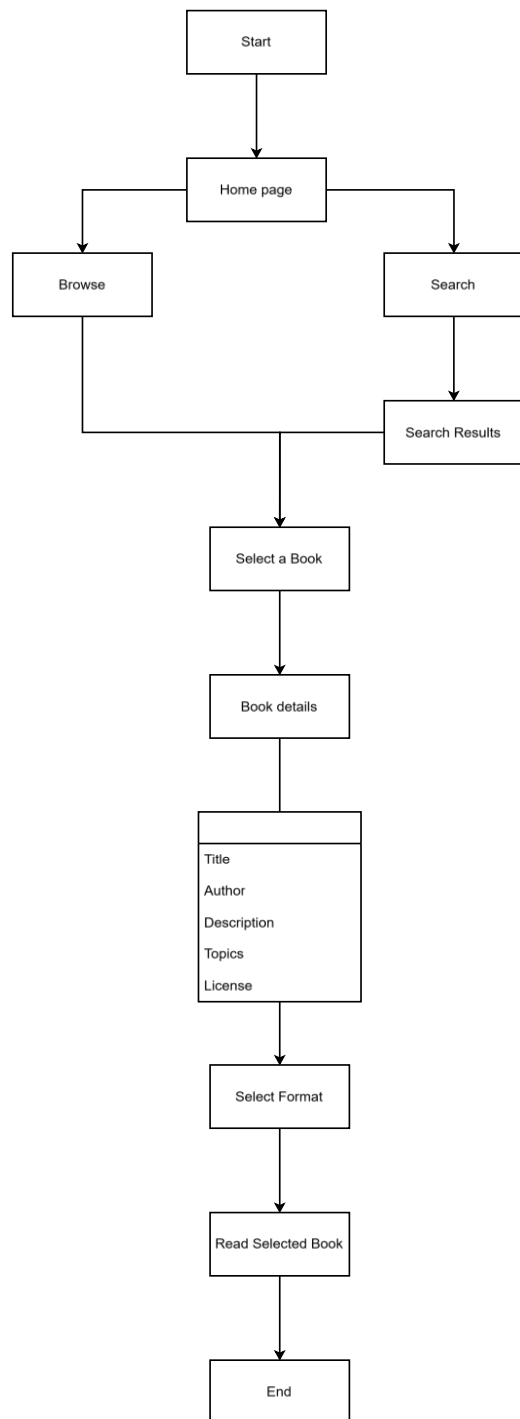


User Flow



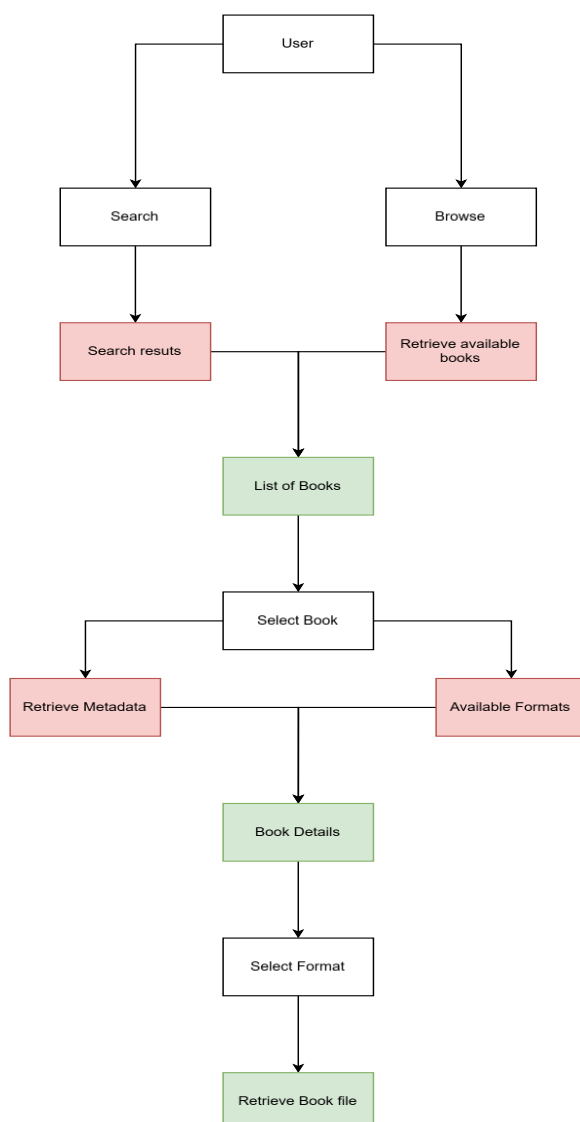
Backend Operations

4 Key Operations

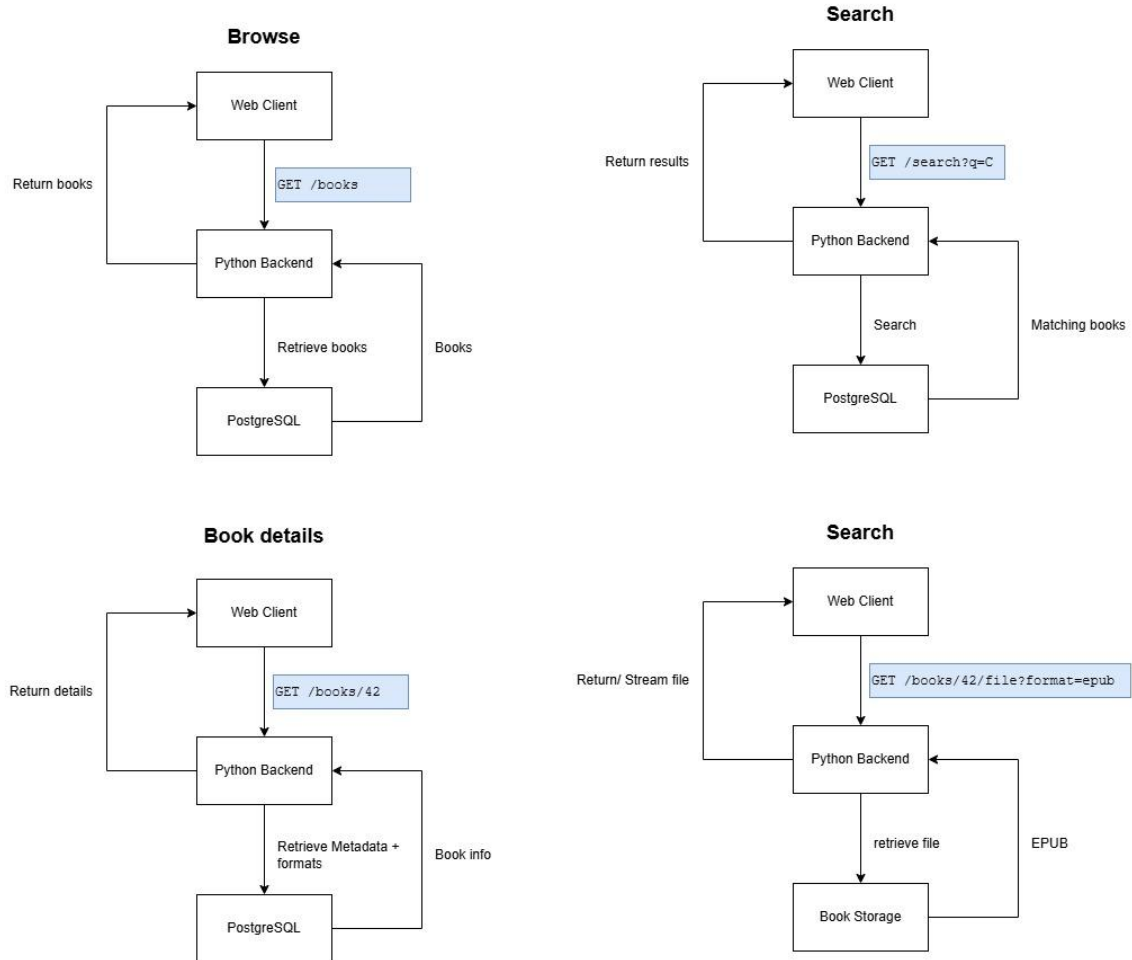
1. Browse Books
2. Search Books
3. See details of a selected book
4. Read the actual book

Each of the operations is a responsibility of the Application's Server

Tracing the responsibility:



Endpoints:



- `GET /books`
- `GET /search?q=...`
- `GET /books/{id}`
- `GET /books/{id}/file?format=....`

Data Requirements

What does Postgres need to contain for the backend to successfully perform its operations?

- Browse needs a collection of books. What Info must each book have?
- Book details need metadata + available format. What info constitutes metadata? What does “available format” mean?
- Search needs searchable information. What fields are searchable?

1) **Browse:** List of books returned

The books card will need something the following data:

- a) ID
- b) Author
- c) Cover
- d) Description (potentially)

2) **Search:** Books matching the query will be returned.

a) Data needed to perform search: What fields should be searchable?

User can search:

- i) By Title
- ii) By Description
- iii) By Author
- iv) By Topic

b) The data needed to display search results to the user:

- i) Same as **Browse**

3) **Selecting a book:** what information should users see about a book?

- a) Book ID
- b) Title
- c) Author(s)
- d) Description
- e) Publication Information
- f) Cover reference
- g) Topics
- h) License
- i) Available formats

4) **Reading a book:**

- a) Book ID
- b) Selected format
- c) Book

Three Categories of data:

1. Information needed to query, filter, search and display (PostgreSQL)
2. Actual book file (File / Object storage)
3. Cover Images (File / Object storage)

Data	Database
Book ID	PostgreSQL
Title	PostgreSQL
Author	PostgreSQL
Description	PostgreSQL
Publication Information	PostgreSQL
Cover reference	PostgreSQL
Topics	PostgreSQL
License	PostgreSQL
Available formats	PostgreSQL
Location	PostgreSQL
Cover Images	File/Object Storage
Actual PDF/EPUB/HTML files	File/Object Storage

How do I organize the data in Postgres?

Most important entity to model is the **Book**:

Potential descriptors:

- 1) ID
- 2) Title
- 3) Description
- 4) Publication Information:
 - a) Publication Year
 - b) Publisher
 - c) Edition / Volume
- 5) Cover reference
- 6) **Author: A separate entity**
A book might have multiple authors, and one author might have multiple books.
What information about the author is needed?
 - a) ID
 - b) Name
- 7) **Topics: Needed for searching and displaying book details. Also model independently.**
A book might have many topics it falls under and a topic might have many books it falls under.
What properties should the topic entity then have?
 - a) ID
 - b) Topic name
- 8) **License: one-to-many relationship and as per data requirements for bookpadi to have a license URL, it would make sense to make it a separate entity.**
 - a) ID
 - b) URL
 - c) License name

9) **Format:** Model as a **separate entity**. Has a many-to-many relationship with the book entity.

a) ID

10) **Location:** Can I store it as a separate entity? If I decide to do this it will have a one-to-many relationship with books, however, I can't store the location alone without something to tell me what format is in that location.

To store location properly, I must establish a relationship between three things: book, format and the location (A location, the book stored there and in what format)

A new entity would be the right choice here

a) Location

b) Book_id

c) Format_id

Serves as the junction of books and format + the location of the book in that format. There won't have to be a junction table for books and format since this table covers that too.

Relationships

Relationship	Book → Entity	Entity → Book
Book → Author	A book must have at least one author	An author must have at least one book association
Book → License	A book must have just one license	A license can exist without any book association
Book → Format	A book must have at least one format	A format can exist without any book association
Book → Topics	A book must have at least one topic	A topic can exist without any book association

ER diagram

